LECTURER: JOHN DOE

REINFORCEMENT LEARNING

REINFORCEMENT LEARNING
TOPIC OUTLINE

Introduction to Reinforcement Learning	1
Dynamic Programming and Markov Decision Processes	2
Bandit Problems	3
Q-Learning	4
Reinforcement Learning Approaches	5
Summary: Reinforcement Learning	6

UNIT 6

SUMMARY: REINFORCEMENT LEARNING

STUDY GOALS



- Recall the key **components of reinforcement learning**, including dynamic programming, Markov decision processes, and bandit problems

- Evaluate the **effectiveness of reinforcement learning** approaches in solving complex decision-making problems

- Understand the principles of **Q-learning** and its role in reinforcement learning algorithms

EXPLAIN SIMPLY



1. What is the fundamental role of dynamic programming in Markov Decision Processes (MDPs) in reinforcement learning

2. How do Bandit Problems apply in real-world scenarios within the context of reinforcement learning?

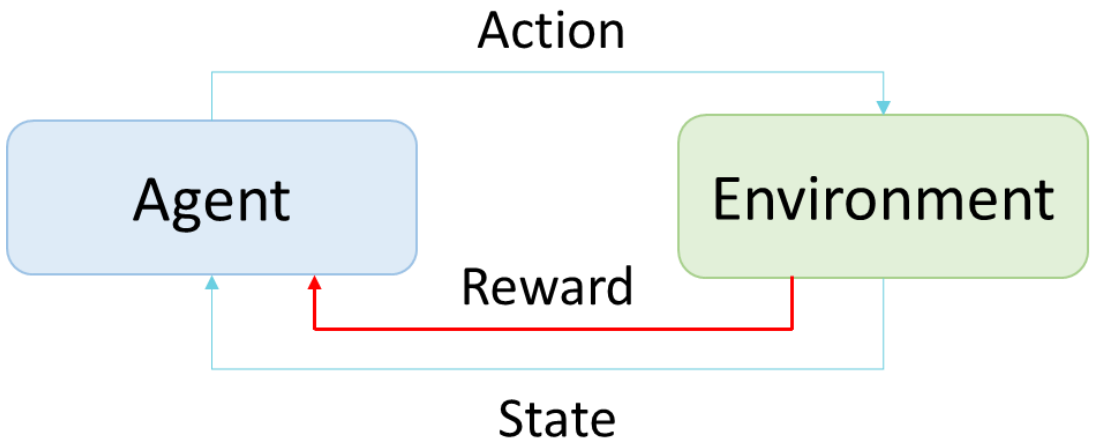
3. When is Q-Learning most effective, and how does it differ from traditional dynamic programming?

REINFORCEMENT LEARNING: THE BIG PICTURE



Optimizing computer decisions through trial-and-error learning

- **State:** current condition
- **Action:** allowed moves in each state
- **Reward:** incentive for actions
- **Policy:** guide decision-making

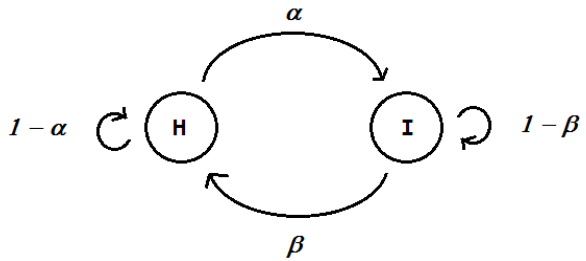


- **Agent**: the learning entity
- **Environment**: external context

Source of the text: Sutton & Barto, 2018.
Source of the image: Own representation.

Markov Decision Process (K, R, P, β)

- State action space K
- Cost function $R(x, a)$
- Probability kernel $P(y \mid x, a)$
- Discount factor β



Bellman Principle of Optimality

- Find and evaluate a policy function to optimize cost accrued
- Current decisions affect future optimality

$$V^*(x) = \min_{a \in X} V^*(x, a) \qquad x \in X$$

Source of the text: Sutton & Barto, 2018.
Source of the image: Own representation.

Value Iteration
Iterative Optimization

- Initialize values for each state
- Update values using Bellman's equation
- Calculate new values based on maximal reward
- Converge to optimal value function

Policy Iteration
The Actor – Critic Method

- Policy evaluation **(Critic)**: evaluate V^{Φ_i}
- Policy improvement **(Actor)**

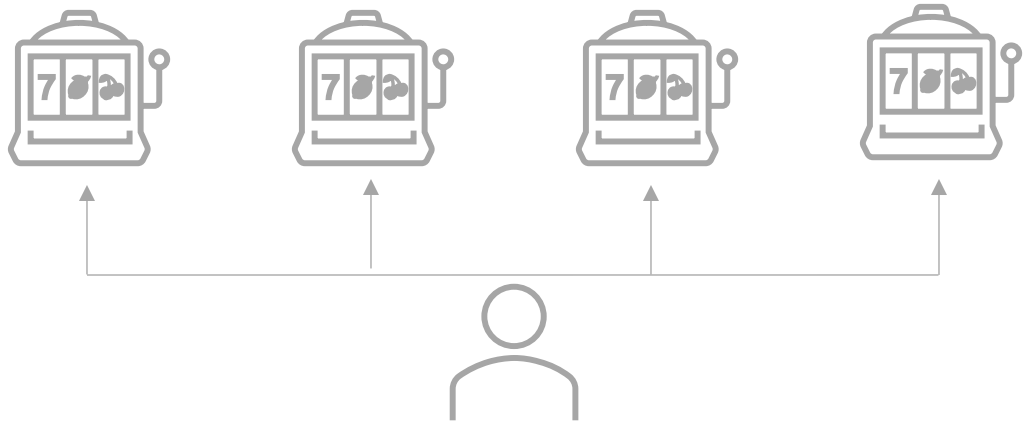
$$\phi_{i+1}(x) = \underset{a \in A_x}{\operatorname{argmin}} R(x,a) + \beta \sum_{y \in X} V^{\phi_i}(y) P(y \mid x, a)$$

- Terminate when optimal policy is found

Source of the text: Sutton & Barto, 2018.

Multiple Arm Bandit: optimize choices among multiple actions to maximize cumulative rewards

- **Multiple Arms:** set of actions
- **Reward Distribution:** uncertain rewards for each action
- **Initial Knowledge**: limited or no reward distribution information
- **Learning**: update action preferences based on observed rewards over time



Maximize rewards:
How to choose actions?

- **Q-factor:** reward when taking optimal action following optimal policy

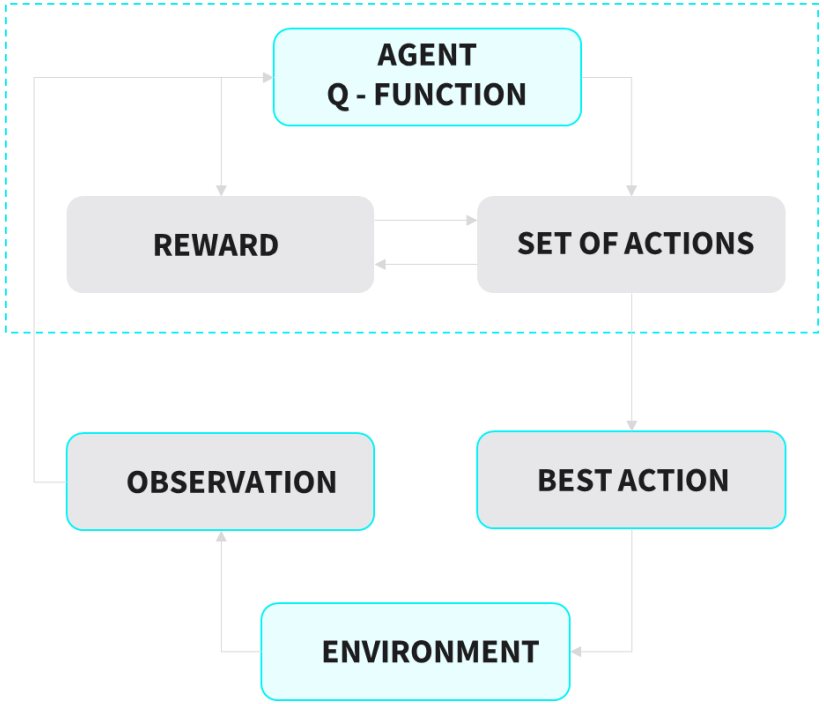
$$Q_1(s,a) \equiv r\,(s,a)$$

$$\pi_1(s) = \underset{\alpha}{\operatorname{argmax}}\, Q_1(s,a)$$

- Convergence assures optimal value function (Bellman equation)

$$V^*(s) = \max_{\alpha}\left[r(s,a) + \gamma \sum_{s'} P(s'\,|s,a)\, V^*(s')\right]$$

Goal: discover optimal action strategy through iterative value updates



Source of the text: Sutton & Barto, 2018.
Source of the image: Own representation.

MONTE CARLO POLICY EVALUATION

Evaluate policies directly from experience

No need to know the structure of the underlying problem



Model Free

1. **Initialize**
   - Policy π to evaluate
   - Arbitrary state-value function V
   - Returns for all states $Returns(s)$
2. **Repeat**
   - Use policy π to generate episode $(s_0, a_0, r_1, \ldots, s_{k-1}, a_{k-1}, r_k)$
   - For each state s in episode
     - Retrieve return r_i for s and append to list of returns
     - Compute value function

$$V(s) = \frac{1}{N} \sum_j Returns(s)_j$$

Source of the text: Sutton & Barto, 2018.

SARSA ALGORITHM

- **Q-learning (off policy)**
  values updated independently of current policy

- **SARSA (on policy)**
  next action follows same policy

1. Initialize $Q(s,\ a)$ arbitrarily
2. Start at initial state s
3. Repeat until terminal state reached
   - Choose action α for state s
   - Take action α, observe reward r, and move to new state s'
   - Choose action a' for new state s'
   - Update Q-values

$$Q(s,a) = Q(s,a) +$$

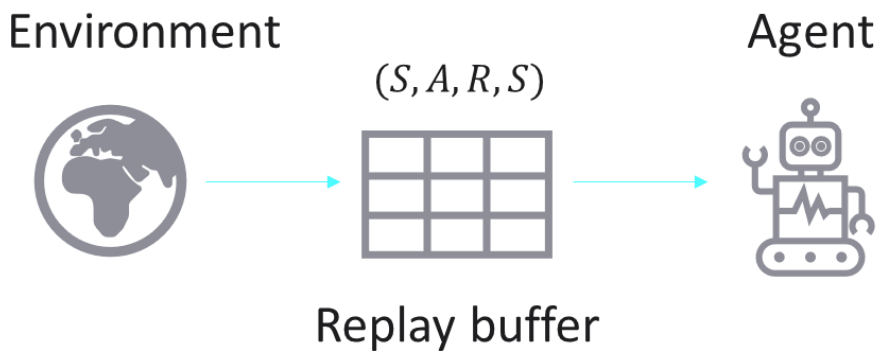
$$\alpha\left[r + \gamma\ \max_{\alpha} Q(s',a') - Q(s,a)\right]$$

4. Set current state to new state $s \mapsto s'$ and current action to new action $\alpha \mapsto \alpha'$

Source of the text: Sutton & Barto, 2018.

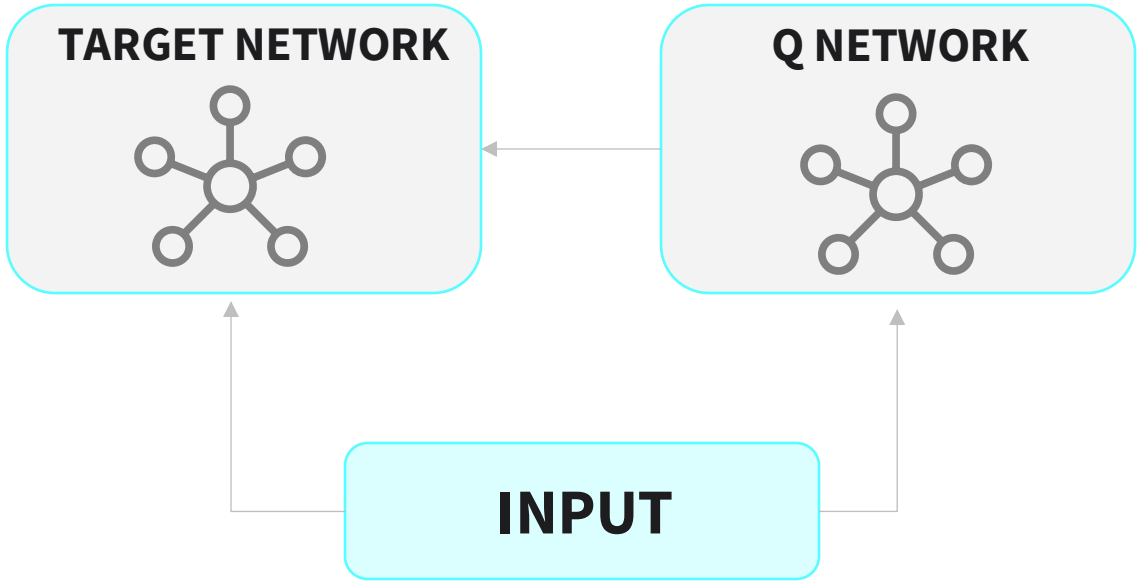
Experience Replay

- Save transitions $(S_t, A_t, R_{t+1}, S_{t+1})$ into buffer and sample batch
- Use batch to train agent



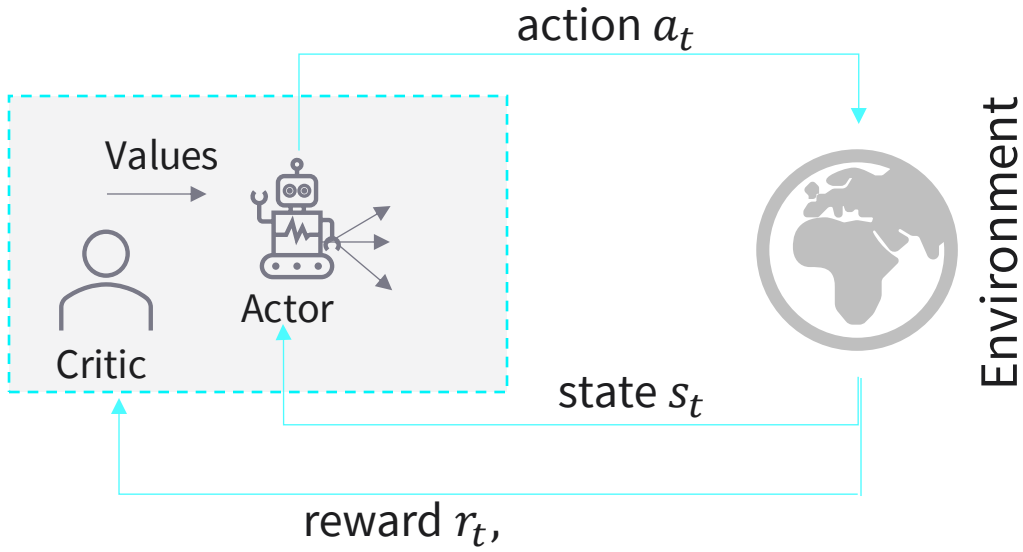
Double Q-Learning

- Utilize separate target and value networks



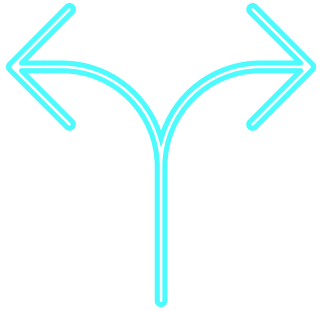
Simultaneous policy evaluation and improvement
Combine policy-based and value-based learning

- **Actor** learn policy (**action selection**)
- **Critic**: estimate value function (**policy evaluation**)
- Enhances stability, convergence, and exploration



Source of the text: Sutton & Barto, 2018.
Source of the image: Own representation.

EXPLORATION VERSUS EXPLOITATION



Exploitation

Should we stick with our known but maybe mediocre results?

- **Greedy:** choose action that gives highest result with current knowledge
- Exploit the environment

Exploration

Or risk something new with unknown rewards?

- ϵ **– Greedy:** introduce small probability for random actions
- Explore the environment

$$\alpha_t = \begin{cases} \alpha_t^* : & with\ probability\ 1 - \epsilon \\ \alpha_t^* : random\ action\ with\ probability\ \epsilon \end{cases}$$

Source of the text: Sutton & Barto, 2018.

REVIEW STUDY GOALS



- Recall the key **components of reinforcement learning**, including dynamic programming, Markov decision processes, and bandit problems

- Evaluate the **effectiveness of reinforcement learning** approaches in solving complex decision-making problems

- Understand the principles of **Q-learning** and its role in reinforcement learning algorithms

SESSION 6

TRANSFER TASK





Case study

You are working on an autonomous delivery robot tasked with efficiently navigating through a complex urban environment to deliver packages. The goal is to optimize the robot's navigation strategy to ensure timely and safe deliveries while considering factors like traffic, obstacles, and energy consumption

Task

Frame this task as an RL problem. How would you optimize the autonomous robot's navigation policy, balancing exploration-exploitation to maximize rewards? Which RL strategies and approaches would you use?

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your results.

The results will be discussed in plenary.





1. A reinforcement learning application is designed to play a card game automatically. A play holds five cards from a standard deck of 52. At each stage the player can discard and replace zero, one or two cards. How many actions are available at each stage
   a) 2
   b) 10
   c) 16
   d) 26



2. Which statement best describes the exploration-exploitation dilemma?
   a) This occurs when the decision maker wishes to both apply the best available policy and collect new data in order to improve the policy
   b) This arises because determination of the optimal policy requires statistical estimation
   c) If the environment changes, the current optimal policy is no longer optimal
   d) In general, a reinforcement learning application must visit every state-action pair infinitely often in order to converge to the optimal policy



2. Which statement best describes the exploration-exploitation dilemma?
   a) This occurs when the decision maker wishes to both apply the best available policy and collect new data in order to improve the policy
   b) This arises because determination of the optimal policy requires statistical estimation
   c) If the environment changes, the current optimal policy is no longer optimal
   d) In general, a reinforcement learning application must visit every state-action pair infinitely often in order to converge to the optimal policy

LIST OF SOURCES

Sutton, R. S., & Barto, A. G. (2018a). Reinforcement learning: An introduction (2nd ed.). MIT press.

© 2022 IU Internationale Hochschule GmbH
This content is protected by copyright. All rights reserved.
This content may not be reproduced and/or electronically edited, duplicated, or distributed in any kind of form without written permission by the IU Internationale Hochschule GmbH.